Eric J. Mei

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EDUCATION

PhD, University of Washington *2023 – present*
Atmospheric and Climate Science, Advisors: Alexander J. Turner, Gregory J. Hakim

MS, Georgia Institute of Technology *2022 – 2023*
Environmental Engineering, Advisor: Armistead (Ted) G. Russell
Thesis: Impacts of Fuel Prices and Regulations on Electricity Generation Emissions and Urban Air Quality

BS, Georgia Institute of Technology *2018 – 2022*
Environmental Engineering, *highest honors*
Minor: Earth and Atmospheric Sciences (Environmental Chemistry)

RESEARCH

Department of Atmospheric and Climate Science, University of Washington *2023 – present*
Graduate Research Assistant (PI: Alexander Turner and Gregory Hakim)

- Developing computationally efficient surrogate models (emulators) for chemistry-climate system
- Performing coupled data assimilation with trained emulators to jointly infer tropospheric composition and physical atmospheric variables

School of Civil and Environmental Engineering, Georgia Institute of Technology *2022 – 2023*
Graduate Research Assistant (PI: Armistead (Ted) Russell)

- Performed thesis research separating impact of natural gas price trends from regulations on air quality
- Sampled ambient mercury at near-road site in joint project with the University of Nevada Reno
- Modeled source contributions to $PM_{2.5}$ composition during Jan-Feb 2022 in Fairbanks, AK

School of Civil and Environmental Engineering, Georgia Institute of Technology *2022*
Research Assistant (PI: Jennifer Kaiser)

- Evaluated existing and novel ethylene oxide (EtO) measurement techniques for Georgia Environmental Protection Division
- Communicated scientific analysis to stakeholders with less technical background in a detailed report

School of Civil and Environmental Engineering, Georgia Institute of Technology *2021 – 2022*
Undergraduate Research Assistant (PI: Xing Xie)

- Identified ability of superabsorbent polymer (PSAP) beads to isolate analytical targets from wastewater
- Helped develop VirusTrack, a commercial method for the use of PSAP beads in COVID-19 surveillance
- Competed in the CEE Entrepreneurial Impact Competition and presented VirusTrack to industry panel

School of Earth and Atmospheric Sciences, Georgia Institute of Technology *2020*
Undergraduate Research Assistant (PI: Yuanzhi Tang)

- Analyzed transformation of contaminants in sewage sludge post-anaerobic digestion
- Obtained N and P concentrations of wastewater samples through spectrophotometry

PUBLICATIONS

8. **Mei, E. J.**, G. J. Hakim, C. Proistosescu, T. K. Bauska, C. Buizert, A. J. Turner (in review). Multidecadal preindustrial methane variability can be explained by noise in the source-sink imbalance. *In review.*

7. **Mei, E. J.**, G. J. Hakim, M. Taniguchi-King, D. Stiller, A. J. Turner (2025). Emulating chemistry-climate dynamics with a linear inverse model. *Atmos. Chem. Phys*. doi.org/10.5194/acp-25-15033-2025

6. Gao Z., **E. J. Mei**, X. He, S. Ebelt, D. Q. Rich, A. G. Russell (2025). Accountability Assessment of Source-Specific Impacts of Regulations on Emissions and Air Quality Using Positive Matrix Factorization. *ACS ES&T*. doi.org/10.1021/acs.est.4c12511

5. Gao Z., **E. J. Mei**, X. He, P. K. Hopke, S. Ebelt, D. Q. Rich, A. G. Russell (2025). Multicity accountability and uncertainty assessment of the impacts of regulations on air quality in Atlanta, New York City, and Southern California. *Atmospheric Environment*. doi.org/10.1016/j.atmosenv.2024.120947

4. **Mei, E. J.**, Z. Gao, P. K. Hopke, S. Ebelt, D. Q. Rich, A. G. Russell (2024). Impacts of fuel prices and regulations on electricity generation emissions and urban air quality. *ACS ES&T Air*. doi.org/10.1021/acsestair.3c00034

3. **Mei, E. J.**, A. C. Moore, and J. Kaiser (2023). Suitability of new and existing ambient ethylene oxide measurement techniques for cancer inhalation risk assessment. *Environmental Pollution.* doi.org/10.1016/j.envpol.2023.122481

2. Gustin, M. S., S. M. Dunham-Cheatham, N. Allen, N. Choma, W. Johnson, S. Lopez, A. G. Russell, **E. J. Mei**, O. Magand, A. Dommergue, T. Elgiar (2023). Observations of the chemistry and concentrations of reactive Hg at locations with different ambient air chemistry. *Science of The Total Environment.* doi.org/10.1016/j.scitotenv.2023.166184

1. Chen, W., **E. J. Mei**, and X. Xie (2022). Virus stabilization with enhanced porous superabsorbent polymer (PSAP) beads for diagnostics and surveillance. *ACS ES&T Water.* doi.org/10.1021/acsest water.2c00239

TALKS

2026 UC Berkeley Cohen Group Meeting, "Multidecadal preindustrial methane variability can be explained by noise in the source-sink imbalance."

2025 AGU Fall Meeting, "Unforced Variability Can Explain Multidecadal to Centennial Methane Variability in Ice Core Records."

2025 Georgia Institute of Technology seminar in Environmental Engineering (invited), "Emulating Chemistry-Climate Dynamics with a Linear Inverse Model."

2025 British Antarctic Survey (invited), "Characteristics of drivers of paleo-methane variability on multidecadal to centennial timescales."

2025 ICECAP Meeting, "Emulating Paleo Chemistry-Climate Dynamics with a Linear Inverse Model."

2024 AGU Fall Meeting, "Discerning Connections between Lifetimes, Modes, Memory, and Timescales in Coupled Chemistry-Climate Models." **Received OSPA (Outstanding Student Presentation Award).**

2023 AQ ATL23 Symposium, "Impacts of Fuel Prices and Regulations on Electricity Generation Emissions and Urban Air Quality." **Awarded Best Talk.**

POSTERS

2026 UC Berkeley Atmospheric Sciences Center (BASC) Symposium, "Emulating chemistry-climate dynamics with a linear inverse model."

2024 PCC Summer Institute, "Lifetimes and feedbacks in the coupled chemistry-climate system."

2024 CS4Env Symposium, "Development of a linear inverse model to emulate chemistry-climate dynamics."

SCHOLARSHIPS, FELLOWSHIPS, and AWARDS

DOD National Defense Science and Engineering Graduate (NDSEG) Fellowship	*2025 – present*
ARCS Foundation Fellowship at University of Washington	*2023 – present*
Georgia Power Fellowship at Georgia Institute of Technology	*2022*
Environmental Engineering and Science Foundation Master's Degree Scholarship	*2022*
Brown & Caldwell LGBTQIA+ Scholarship	*2022*
Georgia Engineering Foundation ACEC Scholarship	*2022*
ADP Henry Taub National Merit Scholarship	*2018 – 2022*
Zell Miller Scholarship	*2018 – 2022*

TEACHING

Department of Atmospheric and Climate Science, University of Washington *2025*
Teaching Assistant – Climate and Climate Change
- Designed and led weekly 1-hour discussion activity sessions

Tutoring and Academic Support, Georgia Institute of Technology *2019 – 2020*
Peer-Led Undergraduate Study Session Leader – Physics 1
- Coordinated with professor to lead study sessions for over 400 students twice a week
- Trained in guiding and facilitating learning for students via CRLA Level II certification

College of Sciences, Georgia Institute of Technology *2019*
Undergraduate Teaching Assistant – Physics 1
- Managed and taught the weekly 3-hour lab activities and exercises of three laboratory sections
- Graded weekly laboratory quizzes and reviewed concepts in succinct sessions

MENTORING

Isabella Showman (University of Washington), *2025 – present*, mentored in research

Coco Lipe (University of Washington), *2024 – present*, mentored in research

Max Taniguchi-King (UC Berkeley), *2024*, mentored in research, co-author on paper, presented a poster at AGU24, now a data analyst at Data Science & Environment at Berkeley

PROFESSIONAL MEMBERSHIPS

American Geophysical Union